

Project 4B - Screenshot Evidence Packet

Aaron Trujillo - Wokwi NTC Thermistor Temperature Test Bench

This PDF packages the screenshots already shared in the project workflow and labels the four image files that should be copied into the GitHub repo screenshots folder.

File name	What it proves
wokwi_ntc_circuit.png	Wokwi circuit wiring: Arduino Uno, NTC sensor, VCC/GND/OUT to A0
wokwi_serial_monitor.png	Serial Monitor output with final CSV columns and FAIL result
python_successful_run.png	Python analyzer ran successfully and generated the report
github_repo.png	GitHub repository is public and project files are uploaded

Important: extracting images from a PDF is annoying. Use the ZIP file I also created. It contains the PNG files already renamed for the screenshots folder.

1. Wokwi NTC Circuit Screenshot

Use this as wokwi_ntc_circuit.png. It shows the Arduino Uno connected to the Wokwi NTC analog temperature sensor and the final FAIL test context.

The screenshot shows the Wokwi IDE interface with a project titled "Project 4B Wokwi NTC Thermistor Temperature Test Bench". The left pane displays the JSON configuration for the circuit, and the right pane shows the simulation of the Arduino Uno board connected to the NTC sensor.

JSON Configuration:

```
1 {
2   "version": 1,
3   "author": "Aaron Trujillo",
4   "editor": "wokwi",
5   "parts": [
6     { "type": "wokwi-arduino-uno", "id": "uno", "top": 0.6, "left": -48.6, "attrs": {} },
7     { "type": "wokwi-ntc-temperature-sensor",
8       "id": "ntc1",
9       "top": -74.2,
10      "left": -39,
11      "attrs": { "temperature": "55", "beta": "3950" }
12    ]
13  ],
14  "connections": [
15    [ "ntc1:GND", "uno:GND_3", "black", [ "h153.6", "v268.8", "h-119.2" ] ],
16    [ "ntc1:VCC", "uno:5V", "red", [ "h144", "v249.6", "h-128.6" ] ],
17    [ "ntc1:OUT", "uno:A0", "green", [ "h134.4", "v238.4" ] ]
18  ],
19  "dependencies": {}
20 }
21 }
```

Simulation Details:

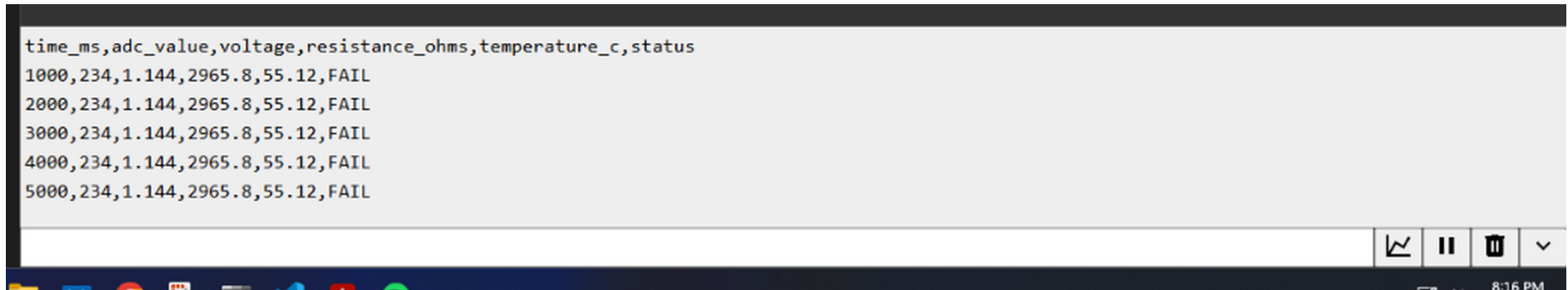
- NTC Temperature Sensor (analog) Temperature: 55.0°C
- Arduino Uno board is shown with the sensor connected to pins A0, 5V, and GND.

Console Output:

time_ms	adc_value	voltage	resistance_ohms	temperature_c	status
1000	234	1.144	2965.8	55.12	FAIL
2000	234	1.144	2965.8	55.12	FAIL
3000	234	1.144	2965.8	55.12	FAIL
4000	234	1.144	2965.8	55.12	FAIL
5000	234	1.144	2965.8	55.12	FAIL

2. Wokwi Serial Monitor Screenshot

Use this as wokwi_serial_monitor.png. It shows the final CSV output columns: time_ms, adc_value, voltage, resistance_ohms, temperature_c, and status.



The screenshot shows the Wokwi Serial Monitor interface. The main area displays CSV output data with the following columns: time_ms, adc_value, voltage, resistance_ohms, temperature_c, and status. The data shows five rows of measurements, all with a status of FAIL. The bottom right corner of the interface contains control buttons: a graph icon, a pause icon, a delete icon, and a dropdown arrow. The system tray at the bottom shows the time as 8:16 PM.

```
time_ms,adc_value,voltage,resistance_ohms,temperature_c,status
1000,234,1.144,2965.8,55.12,FAIL
2000,234,1.144,2965.8,55.12,FAIL
3000,234,1.144,2965.8,55.12,FAIL
4000,234,1.144,2965.8,55.12,FAIL
5000,234,1.144,2965.8,55.12,FAIL
```

3. Python Analyzer Successful Run Screenshot

Use this as python_successful_run.png. This terminal-style image was generated from the successful PowerShell output you pasted in the chat.

Windows PowerShell - Project 4B Analyzer Run

```
(.venv) PS C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench> python
.\python\analyze_ntc_log.py

Project 4B analysis complete.
Input CSV:
C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench\data\ntc_serial_log.csv
Summary:
C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench\reports\project4b_summary.txt
PDF report: C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench\reports\project4b_nt
c_test_report.pdf
Final result: FAIL
Status counts: {'NORMAL': 4, 'WARN': 3, 'FAIL': 4}

(.venv) PS C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

4. GitHub Repository Screenshot

Use this as github_repo.png. It shows the public GitHub repository, folder structure, README, and repo topics based on the screenshot you provided.

The screenshot displays a GitHub repository page for 'project-4b-wokwi-ntc-thermistor-temperature-test-bench' by user 'aarontrujillofz-ee'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main', and there is 1 branch and 0 tags. The repository was last committed 11 minutes ago by 'aarontrujillofz-ee' with the message 'Complete Project 4B NTC thermistor validation report'.

The file list shows the following structure:

File/Folder	Description	Last Commit
data	Complete Project 4B NTC thermistor validation report	11 minutes ago
docs	Complete Project 4B NTC thermistor validation report	11 minutes ago
evidence	Complete Project 4B NTC thermistor validation report	11 minutes ago
plots	Complete Project 4B NTC thermistor validation report	11 minutes ago
python	Complete Project 4B NTC thermistor validation report	11 minutes ago
reports	Complete Project 4B NTC thermistor validation report	11 minutes ago
screenshots	Complete Project 4B NTC thermistor validation report	11 minutes ago
wokwi	Complete Project 4B NTC thermistor validation report	11 minutes ago
.gitattributes	Complete Project 4B NTC thermistor validation report	11 minutes ago
.gitignore	Complete Project 4B NTC thermistor validation report	11 minutes ago
README.md	Complete Project 4B NTC thermistor validation report	11 minutes ago
requirements.txt	Complete Project 4B NTC thermistor validation report	11 minutes ago

The README section is titled 'Project 4B - Wokwi NTC Thermistor Temperature Test Bench'. The 'About' section describes the repository as a 'Wokwi Arduino Uno NTC thermistor temperature validation bench with ADC conversion, Python analysis, plots, and automated PDF reporting.' It lists several topics: python, arduino, embedded-systems, data-analysis, adc, pdf-report, ntc-thermistor, wokwi, hardware-validation, sensor-validation, and csv-logging. The 'Releases' section shows no releases published, and the 'Packages' section shows no packages published. The 'Contributors' section lists 'aarontrujillofz-ee' as the sole contributor.

Copy Steps

Extract the ZIP file and copy these PNG files into:

C:\Users\fall3\Desktop\Project_4B_Wokwi_NTC_Thermistor_Temperature_Test_Bench\screenshots

Then run these commands:

```
dir .\screenshots
git status
git add screenshots/
git commit -m "Add Project 4B validation screenshots"
git push
```

Blunt check: if Git says "nothing to commit," the PNG files are not actually inside the screenshots folder or they were already committed.